

MI-04: Resolution

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Repository status: Solved. The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. [Detailed review and scope](#).

1 MI-04: a characterization of essentially Hermitian off-diagonal blocks

A matrix $X \in \mathbb{M}_n(\mathbb{C})$ is *essentially Hermitian* if $X = \alpha H + \beta I$ for some Hermitian H and complex scalars α, β . The target is the necessity assertion in the following theorem, corresponding to Bourin and Lee's Conjecture 3.3 [1].

Theorem 1.1. *For $X \in \mathbb{M}_n(\mathbb{C})$, the following conditions are equivalent.*

1. *For every positive semidefinite completion $P = \begin{pmatrix} A & X \\ X^* & B \end{pmatrix}$,*

$$\|P\|_\infty \leq \|A + B\|_\infty.$$

2. *The same inequality holds for every such completion whose diagonal-block sum $A + B$ is a scalar multiple of I .*
3. *For every Hermitian T , the largest and smallest eigenvalues of*

$$K_X(T) = \begin{pmatrix} T & X \\ X^* & -T \end{pmatrix}$$

are negatives of one another.

4. *X is essentially Hermitian.*

In particular, no invertibility or distinct-singular-value hypothesis on X is required.

1.1 Scalar-sum completions force extreme spectral symmetry

The implication (1) $\Rightarrow$ (2) is immediate. Assume (2). Fix a Hermitian T and write

$$a = \lambda_{\max}(K_X(T)), \quad b = -\lambda_{\min}(K_X(T)).$$

Since $\text{tr } K_X(T) = 0$, both a and b are nonnegative. The matrix

$$bI_{2n} + K_X(T) = \begin{pmatrix} bI + T & X \\ X^* & bI - T \end{pmatrix}$$

is positive semidefinite, has diagonal-block sum $2bI$, and has norm $a + b$. Thus (2) gives $a \leq b$.

Let $J = \text{diag}(I, -I)$. Direct multiplication shows

$$K_X(-T) = J[-K_X(T)]J^*.$$

Applying the same argument to $-T$ gives $b \leq a$. This proves (3). If “positive” is interpreted as positive definite, use $b + \eta$ in place of b and let $\eta \downarrow 0$; the same conclusion follows.

1.2 A second-order test forces symmetry of off-diagonal magnitudes

Assume (3). It continues to hold with X replaced by εX , for every $\varepsilon > 0$, because

$$K_{\varepsilon X}(T) = \varepsilon K_X(T/\varepsilon).$$

It is also invariant under unitary similarity of X .

Choose any orthonormal basis and write x_{ij} for the entries of X in that basis. Let

$$D = \text{diag}(1, d_2, \dots, d_n), \quad -1 < d_j < 1,$$

and consider

$$K(\varepsilon) = \begin{pmatrix} D & \varepsilon X \\ \varepsilon X^* & -D \end{pmatrix}.$$

At $\varepsilon = 0$, its largest and smallest eigenvalues are the simple eigenvalues 1 and -1 . For small ε , denote their continuations by $\lambda_+(\varepsilon)$ and $\lambda_-(\varepsilon)$. The elementary second-order perturbation formula gives, with $d_1 = 1$,

$$\lambda_+(\varepsilon) = 1 + \varepsilon^2 \sum_{j=1}^n \frac{|x_{1j}|^2}{1 + d_j} + O(\varepsilon^3), \quad (1)$$

$$\lambda_-(\varepsilon) = -1 - \varepsilon^2 \sum_{j=1}^n \frac{|x_{j1}|^2}{1 + d_j} + O(\varepsilon^3). \quad (2)$$

For completeness, if $H_0 u = \lambda_0 u$ is a simple eigenpair of a Hermitian matrix, $\|u\| = 1$, and V is Hermitian, expand an eigenvector of $H_0 + \varepsilon V$ as $u + \varepsilon w + O(\varepsilon^2)$ with $w \perp u$. In an orthonormal eigenbasis (u_j) of H_0 ,

$$w = \sum_{u_j \perp u} \frac{u_j^* V u}{\lambda_0 - \lambda_j} u_j,$$

and the coefficient of ε^2 in the eigenvalue is $u^* V w = \sum_{u_j \perp u} |u_j^* V u|^2 / (\lambda_0 - \lambda_j)$. Applying this to the off-diagonal perturbation gives (1)–(2).

Condition (3) implies $\lambda_+(\varepsilon) + \lambda_-(\varepsilon) = 0$. Comparing the quadratic coefficients and canceling the $j = 1$ terms gives

$$\sum_{j=2}^n \frac{|x_{1j}|^2 - |x_{j1}|^2}{1 + d_j} = 0 \quad \text{for every } (d_2, \dots, d_n) \in (-1, 1)^{n-1}. \quad (3)$$

First take all $d_j = 0$. Then change only d_ℓ to any nonzero number in $(-1, 1)$. Subtracting the two identities proves

$$|x_{1\ell}| = |x_{\ell 1}| \quad (\ell \geq 2).$$

The basis was arbitrary, so

$$|u^* X v| = |v^* X u| \quad \text{for all orthonormal pairs } u, v. \quad (4)$$

1.3 The pair condition implies normality and a collinear spectrum

Complete a unit vector u to an orthonormal basis and sum the squares in (4). Including the equal diagonal terms yields

$$\|X^* u\|^2 = \|X u\|^2.$$

Thus $u^*(XX^* - X^*X)u = 0$ for every u , and X is normal.

Use an orthonormal eigenbasis of X , with eigenvalues $z_1, \dots, z_n$. If $n \leq 2$, these eigenvalues lie on a line. For any triple of indices i, j, k when $n \geq 3$, let $\omega = e^{2\pi i/3}$ and set

$$u = \frac{e_i + e_j + e_k}{\sqrt{3}}, \quad v = \frac{e_i + \omega e_j + \omega^2 e_k}{\sqrt{3}}.$$

These vectors are orthonormal. Equation (4) gives

$$|z_i + \omega z_j + \omega^2 z_k| = |z_i + \omega^2 z_j + \omega z_k|.$$

Writing $a = z_i - z_k$ and $b = z_j - z_k$, the difference of the squared sides is

$$|a + \omega b|^2 - |a + \omega^2 b|^2 = 2\sqrt{3} \operatorname{Im}(a\bar{b}).$$

It follows that every eigenvalue triple is collinear. If two eigenvalues are distinct, all the others lie on the line through those two; if none are distinct, X is scalar. Since X is normal, in either case $X = \alpha H + \beta I$ with H Hermitian. This proves (3) $\Rightarrow$ (4).

1.4 Converse and completion of the equivalence

Assume (4). Multiplying X by a scalar of modulus one changes $K_X(T)$ only by unitary similarity. We may therefore write

$$X = H + ibI, \quad H = H^*, \quad b \in \mathbb{R}.$$

With the unitary matrix

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ iI & -iI \end{pmatrix},$$

direct multiplication gives

$$Q^* K_X(T) Q = \begin{pmatrix} -bI & T - iH \\ T + iH & bI \end{pmatrix}.$$

A singular value decomposition of $T - iH$ reduces this matrix to n blocks of the form $\begin{pmatrix} -b & \sigma_j \\ \sigma_j & b \end{pmatrix}$. Its eigenvalues are therefore the pairs $\pm \sqrt{\sigma_j^2 + b^2}$. In particular, (3) holds.

Finally assume (3) and let $P = \begin{pmatrix} A & X \\ X^* & B \end{pmatrix} \geq 0$. Set $c = \|A + B\|_\infty$. Because $A + B \leq cI$, the matrix

$$P' = \begin{pmatrix} A & X \\ X^* & cI - A \end{pmatrix} = P + \operatorname{diag}(0, cI - A - B)$$

satisfies $P' \geq P \geq 0$. Also

$$P' = \frac{c}{2} I_{2n} + K_X \left(A - \frac{c}{2} I \right).$$

Positivity gives $\lambda_{\min}(K_X(A - cI/2)) \geq -c/2$. By (3), its largest eigenvalue is at most $c/2$. Thus $P' \leq cI_{2n}$ and consequently $P \leq cI_{2n}$. This proves (1) and finishes the proof of Theorem 1.1. $\square$

References

- [1] Jean-Christophe Bourin and Eun-Young Lee. Eigenvalue inequalities for positive block matrices with the inradius of the numerical range. arXiv:2111.15180, 2021. Conjecture 3.3, printed page 6. URL: <https://arxiv.org/abs/2111.15180>.